

SUMMARY

Machine Learning Engineer with expertise in **reinforcement learning**, **large-scale optimization**, and **deep learning**. Skilled in building and deploying **ML systems**, scaling **training pipelines**, and **optimizing inference performance**. Experienced in **bridging** research innovations with production-ready solutions.

TECHNICAL SKILLS

- **ML/AI:** PyTorch, TensorFlow, Transformers, LLMs, LoRA, DDPG, PPO, RLHF, LangChain, Scikit-learn
 - **Systems/Infra:** AWS (SageMaker, Lambda, S3), Spark, Docker, Kubernetes, Git, CI/CD, SLURM
 - **Optimization:** CPLEX, Gurobi, Mosek, OR-Tools, Linear & Integer Programming
 - **Programming:** Python, C, C++, Java, SQL
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EDUCATION

Ph.D. Informatics, (August 2025): **Pennsylvania State University**, University Park, PA

- **Topic:** Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains

M.S. Computer Science, (August 2022): **University of Houston**, Houston, TX

B.S. Computer Science and Engineering, (January 2018): **University of Moratuwa**, Moratuwa, Sri Lanka

EXPERIENCE

Graduate Research Assistant – Penn State University

Aug 2022 – May 2025

- Built transformer-based multi-agent RL system reducing emergency responder reallocation latency **from a couple of minutes to fraction seconds**, scalable to large scale cities (ICML 2024)
- Developed an online vehicle routing system with advance booking; achieved **real-time request confirmation in less a second**, improving service rates against other baselines

Graduate Research Assistant – University of Houston

Sep 2019 – Aug 2022

- Engineered RL solution for transit routing with advance booking that cut **operational costs by 20%**, validated with a partner transit agency (IJCAI 2022)
- Modeled energy-aware scheduling for mixed-fleet buses, reducing annual energy costs by **\$140K** (AAAI 2021)

Software Engineer – LSEG Technology

Jan 2018 – Jul 2019

- Developed and maintained application software using **Java, C++, and Python** with **Object-Oriented principles**, implementing Agile practices (Scrum) to improve productivity, and performance
- Designed and executed **unit testing** and **BDD-based tests** ensuring system reliability, regression compatibility, and enhanced test coverage

Software Engineering Intern – WSO2

Jul 2016 – Dec 2016

- Built an end-to-end **HL7/FHIR monitoring** system using WSO2 ESB, DAS, and BAM for both **real-time (Siddhi)** and **batch (Spark)** analytics of healthcare data
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SELECT PUBLICATIONS

- **A. Sivagnanam** et al.; *Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing*. International Conference on Machine Learning. [ICML 2024]
 - **A. Sivagnanam** et al.; *Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit*. International Joint Conference on Artificial Intelligence. [IJCAI 2022]
 - **A. Sivagnanam** et al.; *Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service*. AAAI Conference on Artificial Intelligence. [AAAI 2021]
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